# Excited State Dynamics on Neutral Argon Cluster

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In the following work we introduce a model to study Excitation Dynamics with Quantum Transition using whole (Diabatised) Diatomics-in-Molecules (Di-DIM) basis. Here, we give an algorithm for coefficient evolution in the basis of time-averaged Hamiltonian to minimise the perturbations. *On-the-fly* adiabatic potential energy curves (PECs) are evaluated using Di-DIM method on which a trajectory is evolved non-adiabatically. The spin-orbit-coupling is included within Di-DIM method. The surface-hopping is employed for hops between adiabatic surfaces. We use this method to study excitation evolution in small neutral argon cluster, ($N = 3, 7, 13$), which has not been done previously. We further investigate the fragmentation channels associated to the relaxation following excitation in the lowest singlet excited state of the 4s band. We observe relaxation to a lower triplet state in $Ar_3^*$, $Ar_7^*$ and $Ar_{13}^*$ clusters. We observe emission of excited atom to be the dominant dissociation channel from a ground state cluster. This is followed by formation of an excimer ($Ar_2^*$) and finally the atomisation. The observation of excimer is dependent on the initial conditions and cluster size. We do not observe formation of a stable excited trimer isomer using Di-DIM. These results are compared with previous studies on ionic systems of similar size and with some experimental observation of cluster and solid Argon.

## I. INTRODUCTION

The fragmentation of rare-gas ionic clusters by electron impact, photoionization, or photoexcitation have been thoroughly studied in the literature. The rare-gas systems have been used to understand the electron motion (in ps regime) in atoms before the recent development in atto-second experiment. With atto-second physics, it is now possible to get better resolution and potentially probe excitation relaxation dynamics on small rare-gas clusters. There is not many reported work on study of excited state dynamics on rare-gas clusters. Argon rare-gas cluster provides an ideal interface between atomic and condensed matter systems.

The majority of theoretical works involving the study of rare gas systems has been limited to ionic clusters. These methods are based on the combination of a DIM model with SO couplings, which is used as *on-the-fly* method for mixed quantum-classical dynamics to model such systems. The results of ionic system studies is summarised in a review article by Bonhommeau et al. [1]. In the previous works to study ionic systems, the $Ar_N^+$ dynamics has been studied by Ehrenfest and surface hopping type dynamics. Ehrenfest dynamics was employed in HWD method (Hemiquantal Dynamics) with whole DIM basis by Amarouche *et al* [2] to study the fragmentation of small argon clusters. Their remark about this type of calculation being close to *ab initio* is unclear. Besides that, the trajectory is propagated on an average potential, i.e., not on true adiabatic surfaces, which makes the assignment of the fragmentation channels rather subjective. The work of Janecek *et al.* [3] used DIM+SO method to calculate the PES. However, the coefficients evolved on adiabatic surfaces. Further work by Janecek *et al.* [4] showed the difference between mean-field dynamics and surface hopping for different rare gas ionic trimers. They reported that mean-field approaches do not give reliable fragmentation results for $Kr_3^+$ and $Xe_3^+$ (except for $Ar_3^+$) when compared to experiments and surface hopping dynamics.

In the current work, we present a theoretical model to study the time evolution of the Rydberg states of excited neutral rare-gas clusters. Our model is based on previous work on ionic clusters. In the following method, we use Di-DIM (Diabatised-Diatomic in molecules) method to evaluate *on-the-fly* potential energy curves. Based on our previous research work [5], we have shown that the excited Argon isomers are not similar to ionic system. it has been speculated however that the excited state dynamics of the two systems would be very similar. In this work we show it is not

the case.

The DIM method is parametrisied using potentials obtained using Hole-Particle-Psudopotential (HPP) and MRCI methods which are diabatised using unitary transformations, as shown in our other work[in progress]. The nuclies are treated classically which are propagated using Beeman's algorithm [6, 7] on the aiabatic potentials. The evolution of the wave packet is based on the method proposed by Janecek *et al* [3] using the whole Di-DIM basis. The transition between the adiabatic states is done using surface-hopping algorithm proposed by Tully [8].

Hence, in the following work we show the importance of diabatisation in the dynamics. The excitation dynamics on excited neutral rare gas systems has not been done, but it was speculated that it would follow the ionic system behaviour. In this work we show that it is not case.

## II. THEORETICAL MODEL

In our previous work we setup the Di-DIM hamiltonian represented as where, the diatomic fragments matrix when represented in a localised pbf basis of atomic functions is given as:

$$\hat{H}_{\text{Di-DIM}} = \begin{pmatrix} h_A^D & h_{AB}^D \\ h_{BA}^D & h_B^D \end{pmatrix} \tag{1}$$

$$h_A^D = \frac{1}{2}\begin{pmatrix} E_{\Pi_{g+u}} & & & \\ & E_{\Pi_{g+u}} & & \\ & & (\alpha)E_{\Sigma^+_{u+g}} & \delta_{\Sigma_{u+g}} \\ & & \delta_{\Sigma_{u+g}} & (\beta)E_{\Sigma^+_{u+g}} \end{pmatrix}; \quad h_{AB}^D = \frac{1}{2}\begin{pmatrix} E_{\Pi_{g-u}} & & & \\ & E_{\Pi_{g-u}} & & \\ & & (\alpha)E_{\Sigma^+_{u-g}} & \delta_{\Sigma_{u-g}} \\ & & \delta_{\Sigma_{u-g}} & (\beta)E_{\Sigma^+_{u-g}} \end{pmatrix} \tag{2}$$

where $\delta_{\Sigma_{u\pm g}}$ are the non-adiabatic coupling terms obtained from diabatization. $E_{\Lambda_{g\pm u}}$ are the energies obtained form the PECs; $\Lambda_{u+g} = \frac{1}{2}(\Lambda_u + \Lambda_g)$ and $\Lambda_{g-u} = \frac{1}{2}(\Lambda_g - \Lambda_u)$ for $\Lambda = \Sigma, \Pi$. Therefore, as an example, $E_{\Sigma^+_{u+g}} = \frac{1}{2}(\Sigma^+_u + \Sigma^+_g)$ and so on. Refer previous work for detailed description [9].

The dynamics method is based on solution of time-dependent Schrodingers Equation using time-dependent Hamiltonian. The calculation of Hamiltonian is done *on-the-fly* for each nuclear configuration using Di-DIM method 1. It is assumed that the Hamiltonian is well defined for any nuclear configuration.

With Hamiltonian being the central point, we start by defining the time-dependent Hamiltonian in perturbation theory as:

$$\hat{H}(t) = \langle \hat{H} \rangle + \delta \hat{H}(t) \tag{3}$$

where $\delta \hat{H}(t) = \hat{H}(t) - \langle \hat{H} \rangle$ is a perturbation to the time-averaged Hamiltonian $\langle \hat{H} \rangle$ . This time independent Hamiltonian is formed in average Hamiltonian theory given by Brinkmann [10].

Now we introduce a stationary orthonormal basis $\sum_\alpha |\alpha\rangle\langle\alpha| = 1$ which diagonalises $\langle \hat{H} \rangle$. The time evolving wave function in Schrodinger's picture in this basis is given by:

$$|\psi_S(t)\rangle = \sum_\alpha C_\alpha(t)|\alpha\rangle \tag{4}$$

We can go from Schrodinger's picture to Interaction picture as using the unitary time-evolution operator $\hat{U}_0(t,t_0) = e^{-i\langle \hat{H} \rangle (t-t_0)}$ and defining by:

$$|\psi_S(t)\rangle = \hat{U}_0(t,t_0)|\psi_I(t)\rangle \tag{5}$$

On applying this in the time-dependent Schrodinger's Equation, we get in the interaction picture:

$$\frac{\partial}{\partial t}|\psi_I(t)\rangle = -i\hat{W}(t)|\psi_I(t)\rangle \tag{6}$$

where $\hat{W}(t) = \hat{U}_0^\dagger(t,t_0)\delta\hat{H}(t)\hat{U}_0(t,t_0)$ is the interaction Hamiltonian.
On solving this differential equation formally we get the time-dependent evolution of wave function as:

$$|\psi_I(t)\rangle = \hat{U}_I(t,t_0)|\psi_I(t_0)\rangle \tag{7}$$

where

$$\hat{U}_I(t,t_0) = \exp\left[-i\int_{t_0}^{t} \hat{W}(t')dt'\right] \tag{8}$$

is the Time-Displacement Operator given by Pechukas *et al* [11] in Feynman's representation [12] of the unitary operator. The unitary time-displacement operator is evaluated using Magnus approximation [13], where we consider the perturbation to be very small for a small time-step $\tau = t - t_0$. This means that in the expansion of the exponential in $\hat{U}_I(t,t_0)$ we neglect all the commutators like, $\left[\hat{W}(t),\hat{W}(t')\right]$, i.e., in first order of approximation $\left[\hat{W}(t),\hat{W}(t')\right] \to 0$. See articles [10, 11].

Using the definition of the wave function $|\psi_I(t_0)\rangle = \hat{U}_0(\tau)\sum_\alpha C_\alpha(t_0)|\alpha\rangle$ and projecting it on the basis which diagonalise $\langle H\rangle$, we get the coefficient evolution as:

$$C_\beta(t) = \sum_\alpha e^{-iE_\beta\tau}\langle\beta|\hat{U}_I(t,t_0)|\alpha\rangle C_\alpha(t_0) \tag{9}$$

where the fast moving phase $e^{-iE_\beta\tau}$ is integrated exactly irrespective of the electronic time-step ($\tau$). On solving in this approximation we evaluate the non-adiabatic couplings and get the time evolution of the coefficients as:

$$C_\beta(t) = \sum_\alpha e^{-iE_\beta\tau} M_{\beta\alpha}(t) C_\alpha(t_0) \tag{10}$$

where $M_{\beta\alpha}(t)$ accounts for the non-adiabatic coupling. All the coefficients $C_\beta(t)$ are then expressed in the diabatic DIM basis because of their continuous nature making the algorithm stable.

The coefficients of the adiabatic state $A$ we wish to follow, $C_A(t)$, are given by projecting the propagated wave packet onto this adiabatic state at time $t$. The projection reads:

$$C_A(t) = \sum_a A_a(t)\sum_\beta U_{a\beta} C_\beta(t) \tag{11}$$

where $A_a(t)$ are the coefficients of the adiabatic state $A$ in the primitive DIM basis and obtained by diagonalization of the hamiltonian at time $t$. The unitary transformation matrix $U_{a\beta}$ is the matrix, which diagonalizes the time-average hamiltonian $\langle H\rangle$, expressed in the primitive DIM basis indexed by $a$.

The Time Averaged Hamiltonian $\langle\hat{H}\rangle$ that enters equation 3 is evaluated. The time averaging has to be done for each matrix element $H_{ab} = \langle a|\hat{H}|b\rangle$, where $|a\rangle$ is the DIM configuration basis, which is expanded on polyatomic basis formed by antisymmetrised product of atomic basis as shown in the PhD manuscript section 3.2 [9]. For better readability, we drop in the following the indexes $a, b$.

We evaluate the Hamiltonian $\hat{H}_0$ at time $t$ using DIM method, then evaluate $\hat{H}_1$ and $\hat{H}_2$ along the trajectory at time $t + dt/2$ and $t + dt$ respectively. Using these three points we solve the 2$^{\text{nd}}$ order polynomial equation $\hat{H}(t) = A + Bt + Ct^2$ to get the coefficients as:

$$\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} 1 & t & t^2 \\ 1 & \left(t+\frac{dt}{2}\right) & \left(t+\frac{dt}{2}\right)^2 \\ 1 & (t+dt) & (t+dt)^2 \end{pmatrix}^{-1} \begin{pmatrix} h_0 \\ h_1 \\ h_2 \end{pmatrix} \tag{12}$$

which gives the time evolving Hamiltonian equation between $t$ and $t+dt$ where $h_0$, $h_1$ and $h_2$ are the matrix elements of $\hat{H}_0$, $\hat{H}_1$ and $\hat{H}_2$ respectively. The time-averaged Hamiltonian is given by;

$$\langle \hat{H} \rangle = \frac{1}{dt} \int_t^{t+dt} \hat{H}(t')dt' \tag{13}$$

Using the previously obtained Hamiltonian $\hat{H}_0$, $\hat{H}_1$ and $\hat{H}_2$, we get the time-averaged Hamiltonian analytically as:

$$\langle \hat{H} \rangle = \left( \frac{\hat{H}_0 + 4\hat{H}_1 + \hat{H}_2}{6} \right) \tag{14}$$

Eventually, the time evolution of each element of the perturbation matrix $\delta\hat{H}(t)$ is approximated by the form $A + Bt + Ct^2 - \langle \hat{H} \rangle$.

With the high density of states in RG clusters, the states are degenerate or close to degenerate which makes the trajectory evolution difficult. Therefore, it is desirable to lift the degeneracy of the states. This is done by using perturbative constraint to rotate the system without changing the energy using a method proposed in the dissertation [14] and work of Zanuttini *et al* [15]. In this method, for a given degenerate subspace D, small perturbative constraint based on the eigenvalues are used to lift the degeneracy while ensuring the continuity of the trajectory. This is done for all the degenerate subspace at each nuclear step.

## III. MODEL VALIDATION

The computation code needs to be validated before using it for physics analysis. For this, we take a minimum, well defined 4*s* excited state PECs for $Ar_2^*$ obtained using *ab initio* MRCI method using MOLCAS.

There are 12 configuration associated to 4*s* excited argon excimer. A 12 configuration space system is difficult to solve. However using symmetry and degeneracy it is possible to reduce this to a 2-state problem which can be solved with simpler algorithm. In this section, we first show the reduction to 2-state system and then using *Mathematica* code we solve the two differential equations for coefficient evolution and compare it with our model. Refer supplementary material, section **??**.

In this demonstration, an Ar atom is placed on the origin while other moves along the z-axis towards the stationary atom. As they come close to each other, the states of each atom are perturbed

leading to change in population via spin-orbit coupling between singlet and triplet states. As we want to evaluate the non-adiabatic molecular dynamics, not the DIM model, the PECs used to parameterise DIM are obtained from MRCI calculation. It is not important to use Di-DIM PECs as the inclusion of $4p$ state makes larger configuration space making the reduction to 2 state system complicated. The Hamiltonian elements $a(t)$, $c(t)$ and $k$ in equation 15 are obtained from the DIM Hamiltonian. The PEC are $a(t) = {}^3\Pi_u^+(t) + \kappa$, where $\kappa = 477.193$ cm$^{-1}$ is the spin-orbit constant, $c(t) = {}^1\Sigma_u^+(t)$ and $k = \sqrt{2}\kappa$.

The PECs obtained for these colliding dimer are obtained in dynamics for $10^5$ a.u. (200 fs) with a time-step of 0.5 a.u. (12.05 as) in our model. As this method is sensitive to the time-step especially in the region of crossings, we determine that the maximum time-step which can reliably used in dynamics is 2 a.u. in our model.

The potential energy curves a(R) and c(R) crosses and the coupling $\kappa\sqrt{2}$ is independent of time i.e., on the distance R. Now the diagonalization gives the adiabatic states and we can solve the TDSE to get the coefficient evolution of this two state system as:

$$i\frac{d}{dt}\begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} = \begin{pmatrix} a(t) & k \\ k & c(t) \end{pmatrix}\begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \tag{15}$$

where $a(t) = a(R)$ and $c(t) = c(R)$ are the functions obtained by spline interpolation of PECs and $k = \kappa\sqrt{2}$.

This test system is a collision between two neutral argon atom in excited state. The distribution of initial excitation on the adiabatic states is; $|c_1(t_0)|^2 = 0.98$ and $|c_2(t_0)|^2 = 0.02$ with real amplitudes. With no hopping, the exchange of population only comes via non-avoided crossings. With the intention to validate the model, we look at the adiabatic and diabatic population evolution, non-diabatic couplings and probability of hop.

Figure 1 shows the comparison of the results obtained using *Mathematica* [16] by solving equation 15 and our model. The figure a) and b) in 1 shows respectively the adiabatic and diabatic population evolution using *Mathematica* and model. The propagation with Mathematica for the diabatic states with fast phase oscillations matches perfectly with our numerical integration. As a result, the adiabatic population evolution as shown in figure (a) matches perfectly for the two methods. However, due to state crossing we see the population evolution across an non-avoided crossing in a2 by the change of colour. With a larger time-step, these fast oscillations are not

integrated exactly leading to vulnerabilities in the calculation and eventually to the result. It is also worth mentioning that the this system is very sensitive to the initial conditions.

Further, the non-adiabatic couplings and probability of hop from the current state to the adiabatically coupled state is also shown. The non-adiabatic couplings are in good agreement in both the models. However, the probability of jump has some discrepancies. The difference between the two is due to state switching which can be seen in change in colour from blue to orange and back.

## IV. DISCUSSION

The following presents the Excitation Transfer as a function of collision energy, which can happen for a trimer is a large cluster. In the simulation, we also want to see the decay time of Ar3* and excitation-localisation dynamics. From HPP, we learned that the Ar3* is actually Ar2*+Ar, with the lowest isomer to be in co-linear geometry.

### A. $Ar_3^*$ Dynamics

**Excitation transfer by Collision Dynamics:** We have explored two collision processes for the trimer in co-linear geometry. In the first case, an excimer collides with a ground state atom and, in the second case, an excited atom collides with a ground state dimer. This is a test simulation, which is difficult to be done experimentally, however slow collision processes can be thought of as a part of excitation decay and transfer in a cluster where the excitation decay is initiated by the collision. The time scale for the simulations is between $10^5$ to $5 \times 10^5$ (600-800 fs) with a time-step of 0.5 au (12.5 as). It is important to take a small time-step for proper evaluation of overlap.

CASE 1: We start with a stationary lowest excited state excimer and a ground state atom approaching with energy between xx and yy ..... velocity of $2.74 \times 10^{-3}$ au (1.48eV). 10 trajectories are run from this geometry and we observe the excitation transfer from the excimer to the single atom. The excitation transfer is dependent on the kinetic energy of the collision. In this system, we have high kinetic energy. As we see, after the collision, we observe atomisation. The following figure shows few example PEC and motion of ions associated to this collision process. We see in

first example, with high kinetic energy, it is possible to go to a higher excited state. In the other example, we see the excitation being decayed to a lower triplet state.

CASE 2: A ground state dimer is collided with an exited argon atom with initial velocity of $1.23 \times 10^{-3}$ au (0.66eV). The initial kinetic energy in this system is lower than the previous case and we can refer to this as slow collision process, with transfer of excitation from the atom to the dimer. This trajectory has a single jump during collision and has a possibility to decay to a lower-energy state, as observed in our discussion of the two-state model in chapter 6. We observe the decay and transfer of excitation from the colliding atom to the atom on the other side leading to formation of two ground state atoms. These 2 atoms have some kinetic energy and might not bind together as a ground state dimer.

**Fragmentation Dynamics:** The fragmentation dynamics of the trimer is done starting at a ground state geometry. The cluster is then promoted suddenly to one of the lowest-energy singlet exited state of Ar. There are 16 excited states including the *ad hoc* 4p-states for each atom, making a total of 48 states for $Ar_3^*$. The choice of the lowest singlet state is not perfectly well defined due to spin-orbit coupling. We consider the starting state #29, which has a 75% of singlet character. The dynamics is followed during 5 ps. We evolve 30 trajectories on the ground state geometry which is thermalised at T=20K for 5 ps initially. This gives possible random initial kinetic energy and position to each atom for these trajectories. We observe the relaxation to the lower triplet states by non-adiabatic couplings associated with spin-orbit couplings.

In Figure 2, we give examples of evolution of the system on adiabatic PECs which are presented as black bold and highlighted yellow. We observe 100% of the trajectories leading to atomisation in all the cases. The timescale of dissociation is different due to initial kinetic energies imparted to the atoms. We also observe the decay of excitation to lowest triplet excited state which is accompanied by formation of an excimer. However, as the energy in the system is much higher than excimer binding energy, and it atomises. To supplement the dynamical study on state 29, in few of the trajectories we started from a higher singlet state 35. In this case also observed the atomisation of Ar as well but the trajectories do not relax to a lower triplet state. Unlike ionised clusters, excited neutral clusters are highly unstable. It would be interesting to have some experimental observations to ascertain this conclusion.

## B. $Ar_7^*$ Dynamics

### 1. Dissociation Dynamics

The excitation dynamics of $Ar_7^*$ clusters start at the ground-state geometry, and argon atoms are given an initial kinetic energy of 30 K. There are 16 excited states including the *ad hoc* 4p-states in the atomic basis set, making a total of 112 states for $Ar_7^*$. We assume instantaneous excitation at this geometry into one of the lowest singlet excited state. This is the lowest-energy singlet state with a singlet character of at least 80%. Initially the excitation is broadly delocalized over the whole cluster. The excitation localisation is also observed to be dependent on the initial geometry. For instance, if couple of atoms are close to each other after the initial thermalisation, they will carry more excitation and if a cluster is formed with a separated atom, the separated atom is observed to have the excitation localisation on that single atom. The dynamics lasts for 5-10 ps with a time step of 0.5 au (12.5 as) or 1 au. In the following dynamics, we have 40 trajectories distributed over different types of dissociation channels.

**Atomisation:** We start with the channel in which all the atoms blow apart, and is given by:

$$Ar_7^* \rightarrow Ar^* + 6(Ar) \tag{16}$$

This implies that the excitation is carried by a single atom without formation of any cluster. We observe 21% of the trajectories (8 out of 38) in this channel. We do not expect this percentage to increase significantly with increasing cluster size due to increasing cohesion energy of the ground state atoms. Some of the trajectories decay rapidly ($<$ 3ps) to a lower triplet state associated to $J = 0$ or 1. An example is shown in Figure 3a. Access to low-energy triplet states with $J = 2$ is observed in only one of the trajectory, which is shown in the Figure 3b. In this trajectory, as excitation relaxation happens on a cluster of 5 atoms at 4 ps, the atoms gain kinetic energy which is greater than the binding energy of the system, leading to atomisation. This trajectory has few more hops on its way to dissociation. All this happens within a short time of 1 ps. This shows that, as the system gains kinetic energy, the relaxation dynamics becomes faster. Atomisation is not expected to be a common dissociation process for larger clusters. The other channels associated to evaporation of ground or excited atoms or clusters are more favourable. This is consistent with the sputtering experiment on large clusters, where they interpret their observation of luminescence

as due to emission of excited atoms and excimer [17].

**Excited Atom Emission with Neutral Cluster Formation**: In this dissociation channel, we look for the emission of an excited atom with formation of a ground state cluster. We can observe the formation of $Ar_2$, $Ar_3$, $Ar_4$ and $Ar_5$, which are given as;

$$Ar_7^* \rightarrow (Ar^*) + (Ar_2) + 4(Ar) \qquad 32\% \qquad (17)$$

$$Ar_7^* \rightarrow (Ar^*) + (Ar_3) + 3(Ar) \qquad 32\% \qquad (18)$$

$$Ar_7^* \rightarrow (Ar^*) + (Ar_4) + 2(Ar) \qquad (19)$$

$$Ar_7^* \rightarrow (Ar^*) + 2(Ar_2) + 2(Ar) \qquad (20)$$

$$Ar_7^* \rightarrow (Ar^*) + (Ar_5) + (Ar) \qquad (21)$$

We observe 36% gathering dissociation channels where larger clusters are produced.

As an example, in Figure 4, we show the dynamics leading to the formation of a ground state dimer with an evaporating excited atom. Starting from one of the lowest singlet state, the trajectory evolves on a mixture of singlet-triplet states, which then decays into a triplet state. In Figure 4a, we see the relaxation of the adiabatic state to a $J = 0$ triplet state followed by a hop at 1.7 ps. In Figure 4b we observe several hops between 1.7-1.8 ps leading to $J = 2$ triplet state. Following relaxation, kinetic energy is gained by the excited atom, which then evaporates along ground state atoms. In Figure 4a we have slow and small amplitude oscillations, while these oscillations are faster and of larger amplitude in Figure 4b. The difference between the two figures is due to the difference in their internal energy. In Figure 4a the system is on excited state 61 and the kinetic energy release in the system is small whereas, in Figure 4b the system is on excited state 18 and the kinetic energy release in this system is higher leading to a faster vibrating dimer. In Figure 4b, following the adiabatic state after the hop, we see negligible oscillations of the current adiabatic state, highlighted in yellow. The small oscillations in the yellow PEC are due to the electronic states of the well separated oscillating ground state dimer while the excitation is carried by one of the leaving argon. For some of the non-followed states (not highlighted in yellow), the excitation is localised on the vibrating dimer, producing slightly larger amplitude in the potential energy periodic variation. We also identify the other excited PECs of $Ar^*$, highlighted in red at dissociation limit, which are almost parallel to the currently followed adiabatic state after 2.5 ps, during which the excited atom has separated enough and do not interact anymore with the ground

state dimer. The PECs of $Ar^*$ beyond 3 ps show the fine structure splitting of triplets for different $J = 0, 1$ and 2 and the singlet states. These PECs also show the excellent treatment of degeneracies in our algorithm.

In other situations associated with the formation of larger isolated clusters, we also observe, following relaxation, the excitation to be predominantly localised on an atom. The rearrangements of the cluster are a consequence of its internal energy. This internal energy may lead to the evaporation of several ground state atoms and to the formation of a smaller ground state cluster. The excess energy can also lead to the formation of several neutral clusters, like two argon dimers. This effect becomes significant when the cluster size increases.

**Excimer Evaporation:** This channel is associated to the evaporation or separation of an excimer ($Ar_2^*$) from other atoms in ground state. These ground state atoms can either be dissociating individually or in a form of ground state clusters. We found 16% trajectories forming an excimer together with ground-state atoms, produced by atomization, or a ground state cluster. The emission of the excimer with other ground state atoms is similar to atomisation but with an excitation located on the dimer, given by;

$$Ar_7^* \rightarrow (Ar_2^*) + (Ar) + 4(Ar) \tag{22}$$

$$Ar_7^* \rightarrow (Ar_2^*) + (Ar_{5-l}) + l(Ar) \tag{23}$$

Unfortunately, we do not observe excimer associated to the formation of the lowest triplet excited dimer. All the observed excimers are in a high excited state. For most of the trajectories, the dynamics is probably not long enough for them to have relaxed into lower triplet state. Moreover, such a relaxation is likely to break the excimer, thus leading to one of the excited atom channel.

The emission of excimer with formation of a ground state cluster is barely observed, but we observe few trajectories where a ground state cluster is formed alongside an excimer. However, these trajectories are in highly excited states. Nevertheless, this shows the possibility for the formation of ground-state clusters of variable sizes when increasing the system size. It is important to note that, in the interpretation of the experimental observations, they do not report the excitation to be on three atoms.

The identification of such a trimer is possibly quite difficult experimentally.

### 2. *Diabatisation Effect*

To understand the effect of diabatisation, we explored the same $Ar_7^*$ system while using the dimer PECs which are not diabatised. Instead, the four lowest-energy PECs built by the H-DIM method for each spin symmetry have been considered. This is associated to H-DIM method used to evaluate *on-the-fly* PES for trajectory evolution. To perform this dynamics, we used 10 trajectories with same initial conditions for the sake of consistency. However, the starting state is not the same since the atomic configuration space is reduced from 16 to 12 as there are no higher ($4p$) state included for each atom. The time limit of the calculation is 5 ps with a time step of 0.5 au. Though the dissociation channels are similar to Di-DIM simulations, we observed different dynamics. The major difference comes from the possibility for a sharing of the excitation among 3 atoms, which is consistent with the conclusions reported by Naumkin and Wales, who explored the lowest-energy triplet PES for the isomer of minimum energy. We did not perform enough statistics to report a reliable distribution of fragmentation channels at the moment. This calculation was done to get the flavour of the effect of diabatisation on the dynamics. This obviously needs to be investigated in more detail.

Based on our limited observation, we found that the excitation localisation using this method leads to the formation of an excited atom or dimer, but the excitation is more delocalised. The excitation does not localize rapidly on an excimer or an atom as the system evolves. This leads to an excitation being distributed over a greater number of atoms than for Di-DIM simulations. In the dissociation channel, though the excitation is located on a single atom, almost 80% of excitation is spread over other adiabatic states. For example, in one of the trajectories, we observe the excitation to be on state 40, which is a triplet state, localised on a single atom. Though all the excitation in this state is being carried by this single atom, the value of un-normalised $p$ is only 21% of the starting population is in this state. This means that the population does not tend to concentrate on a single adiabatic, as for Di-DIM.

### C. Summary

With the $Ar_7^*$ relaxation dynamics we explored several dissociation channels, which were previously unavailable in trimer simulation, then extending our study of the argon trimer. In this dynamics simulations, the main dissociation channels are the emission of either an excited atom , the formation of an excimer or both of them. They can either be accompanied by atomising ground state atoms or formation of smaller ground state clusters or both. We were able to explore all these cases. The results are presented in table I. 84% of trajectories leads to the emission of excited argon atom and this is a mixture of sub-channels and different types of atoms. Among these 84%, the complete atomisation of the cluster represent 21% of the trajectories. The emission of excited atoms accompanied by ground-state clusters composed of 2 to 5 Ar atoms is also found. This represents the majority of trajectories.

The remaining 15% of the trajectories correspond to the formation of an excimer. In most of the trajectories, the excimer is emitted with other ground state atoms. However, the system does not decay to lower-energy triplet states after 5 ps.

It is also instructive to explore the relaxation by having just singlet states in the system. Thus, we removed the possibility for singlet and triplet mixing. The PES are slightly different and the hopping probability among the adiabatic states is reduced. By keeping only singlet states, we observed that the system atomizes with excitation localising on one of the dissociating atom. Since we only have non-adiabatic non-adiabatic couplings and not the spin-orbit couplings, fewer hops occur. When a non-adiabatic hop occurs, it imparts kinetic energy which leads to atomisation, as the trajectories are not binding in high singlet state. By starting lower-energy excited state, the system relaxes to a lower-energy state but not to the lowest- energy excited state within the 10 ps in the dynamics. This implies that the relaxation dynamics is slower without spin-orbit coupling. Starting on a lower singlet state, we observe the formation of some ground state clusters but no bound excimers. Using pure triplet states instead of singlet states in the simulations yields similar conclusions. These results are similar to the case of full configuration basis including both singlet and triplet states.

The system starts close to its ground state geometry with some initial random kinetic energy. The corresponding atomic momenta are not sufficient to enable efficient non-adiabatic dynamics.

Therefore, the population spreads among many nearly degenerate states, but we do not observe the energy relaxation towards adiabatic states significantly lower in energy. If we increase the temperature of the system, it might be possible to observe more hops and thus faster relaxation.

These simulations are qualitative and give us the flavour of excitation dynamics using our algorithm. Nevertheless, even in this minimum simulation, we modeled dissociation processes observed in sputtering experiments. We observe the localisation of excitation on the excimer which was previously not done. We also proved the possibility for evaporation of excited atoms and formation of excimers.

## V. OUTLOOK

Using Di-DIM with spin-orbit couplings, we performed the mixed quantum-classical dynamics of clusters initially prepared in their ground state geometry and excited in the lowest-energy singlet excited state. The wave-packet propagation is achieved within the interaction picture. We introduced improvements to the propagation algorithm by reducing the perturbation using the average Hamiltonian approach. In doing so, we eliminated the fast electronic oscillations in the wave-packet propagation as we integrated analytically over a small time step. Small time-step and small perturbations lead to smooth evolution of the adiabatic population over the nuclear motion, without losing the phase. The coefficients are propagated in an arbitrary basis-set for each nuclear time-step before being projected to get the adiabatic population of the present state. The nuclei propagation uses the Beeman algorithm. The forces are calculated analytically. A simple state flipping algorithm is used, which evaluates self-overlap to follow the desired trajectory across non-avoided crossings. Fewest Switches Surface Hopping (FSSH) algorithm is used for avoiding crossings. Using this algorithm, we performed long dynamics, usually 10ps, despite a small time-step. Typically, the time-step used is 10as. This time step can be further increased or reduced with the possibility of making the dynamics faster or slower depending on the system.

This propagation algorithm is validated by a set of separate calculations performed on *Mathematica* [16] for $Ar_2*$. For such a cluster, it was possible to reduce the 12-state problem to a 2-state problem, a singlet and a triplet with spin-orbit couplings. The agreement between our model and Mathematica calculations for small time steps is remarkable. Based on this validation, we did pre-

liminary dynamics on small-size clusters to check for the stability of our algorithm. We found that it is essential to keep the time-step small to avoid artefacts when integrating differential equations. We successfully showed the transfer of excitation and its dependence on the kinetic energy in the trimer system. We also showed the relaxation and transfer of excitation from a single excited atom to the ground state dimer leading to an excimer.

We also observed that, for a trimer at its ground state geometry prepared in its lowest singlet excited state, there is complete dissociation due to an excess of kinetic energy. This shows that a high excited state can not exist for a long time in a trimer and as it relaxes to a lower triplet state, the kinetic energy gained by the atoms is much higher than the binding energy of the excimer. Over a time scale of 5 ps, we observed short ($< 1$ ps) and medium ($< 3$ ps) lived trajectories depending on their initial linear momentum, however, they all atomise before 3 ps.

To explore different dissociation channels we simulated the excitation relaxation in the $Ar_7^*$ cluster. We started from their ground state geometry and prepared the cluster in its lowest singlet excited state, mimicking a cluster obtained by photo-excitation. Similar to $Ar_3^*$, this is not a pure singlet state. The relaxation of this system gives a deeper insight into the dynamical process involved and opens the way for exploring other dissociation channels. We observe three major channels among 38 trajectories; 1) complete atomisation of the ground state cluster noted in 21% of the trajectories, 2) separation or emission of an excimer from the ground state cluster is observed in 15% trajectories and 3) production of an excited atom together with a ground state cluster is the main dissociation channel and occurs in 66% of the trajectories. The main relaxation channel is thus the formation of $Ar^*$ (85%), which dominates the observation of $Ar_2^*$ (15%). We observed the formation of larger excited clusters in only a couple of trajectories in our 5-10 ps dynamics. They are unstable as they are in highly excited states, which would eventually lead to the formation of an excimer with emission of ground-state atoms or to the formation of small ground-state clusters with emission of excited or ground-state atoms. Therefore, the energy brought into the cluster by excitation is too large to maintain the cohesion of more than 2 atoms for the small clusters (N=7) investigated here. Hence, we do not observe any stable clusters bigger than a dimer carrying the excitation after relaxation. For a trajectory living in a highly excited state on a small cluster, the atomisation of the cluster is imperative.

In this work, we showed that excitation relaxation leading to dissociation can happen at the

beginning of the dynamics or can last for more than 10ps depending on the initial motion of the nuclei. We find that the 5-10 ps time scale is insufficient for the relaxation of the current followed adiabatic states towards the lower energy states due to weak non-adiabatic couplings in the system coming from the kinetic coupling.

Even for small clusters like $Ar_7^*$, the density of states is high, and there are many degenerate or nearly degenerate adiabatic states. Due to this high degeneracy, we observe the distribution of population among many adiabatic states in the course of relaxation.

In our minimum dynamics on a small cluster ($N = 7$), we noted the localisation of the excitation on an excimer or an atom in the beginning of the dynamics, i.e. within first few picoseconds. This observation is consistent with a previous experimental interpretation given by Moller *et al.* [**? ? ?** ] and also by Lietard *et al.* [17] for Argon clusters. This is also consistent with the experimental work on solid argon by Riemann *et. al* [**? ?** ], who reported the formation of these excited species. They do not give any information about $Ar_3^*$ or any other bigger excited cluster, which is probably very difficult to observe as excitation localises rapidly on an atom or an excimer.

We also illustrated the role played by diabatisation in dynamics by comparing the H-DIM and Di-DIM model dynamics results for $Ar_7^*$ with the same initial conditions. Investigating $Ar_7*$ clusters, we found that the dissociation channels for dynamics using both models were similar. However, we observed the excitation localisation on three atoms when using the H-DIM model, which is not observed in the Di-DIM model. In H-DIM, the formation of an excimer is scarce and we might expect, for larger parent clusters, that proportion of excimer or bigger excited clusters would be much lower in H-DIM than in Di-DIM. Thus, the nature and the number of states used to parametrise the DIM model influences the outcome of the relaxation dynamics.

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TABLE I: Distribution of $Ar_7^*$ dissociation channels. 38 trajectories using Di-DIM parametrisation have been run.

| Dissociation Channel | | Number or trajectories | Total |
|---|---|---|---|
| Atomisation | $Ar^* + 6Ar$ | 8 | |
| Excited Atom: 1 | $Ar^* + Ar_2 + 4Ar$ | 7 | |
| Excited Atom: 2 | $Ar^* + Ar_3 + 3Ar$ | 7 | |
| Excited Atom: 3 | $Ar^* + Ar_4 + 2Ar$ | 5 | |
| Excited Atom: 4 | $Ar^* + Ar_5 + Ar$ | 4 | |
| Excited Atom: 5 | $Ar^* + 2Ar_2 + Ar$ | 1 | 32 |
| Excimer: 1 | $Ar_2^* + 5Ar$ | 2 | |
| Excimer: 2 | $Ar_2^* + Ar_4 + Ar$ | 2 | |
| Excimer: 3 | $Ar_2^* + Ar_3 + 2Ar$ | 2 | 6 |

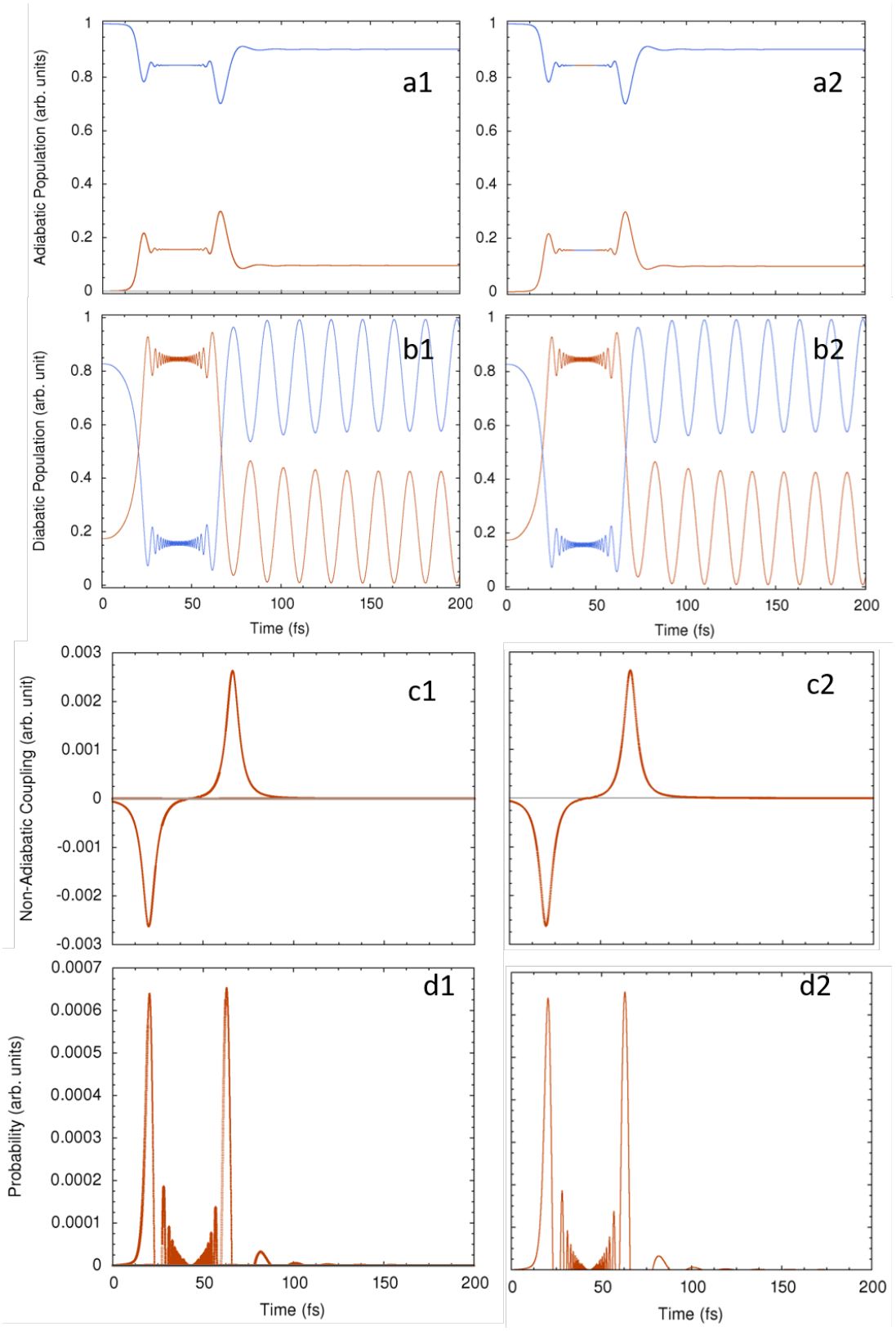



FIG. 1: Validation using Mathematica

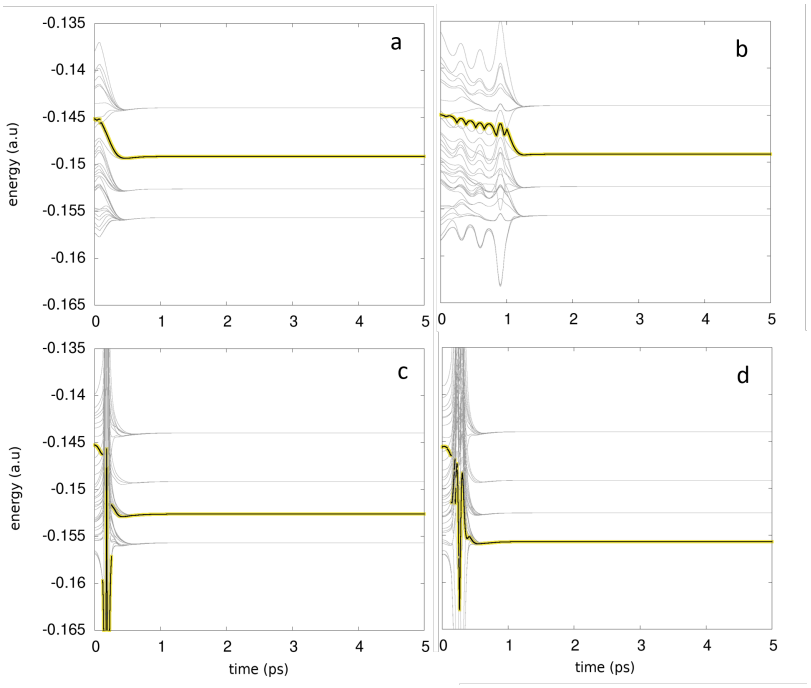



FIG. 2: Figures (a-d) shows the PEC and trajectory evolution on some of the ground state geometry from the lowest singlet excited state. c and d shows the relaxation of excitation to form excimer which eventually dissociate.

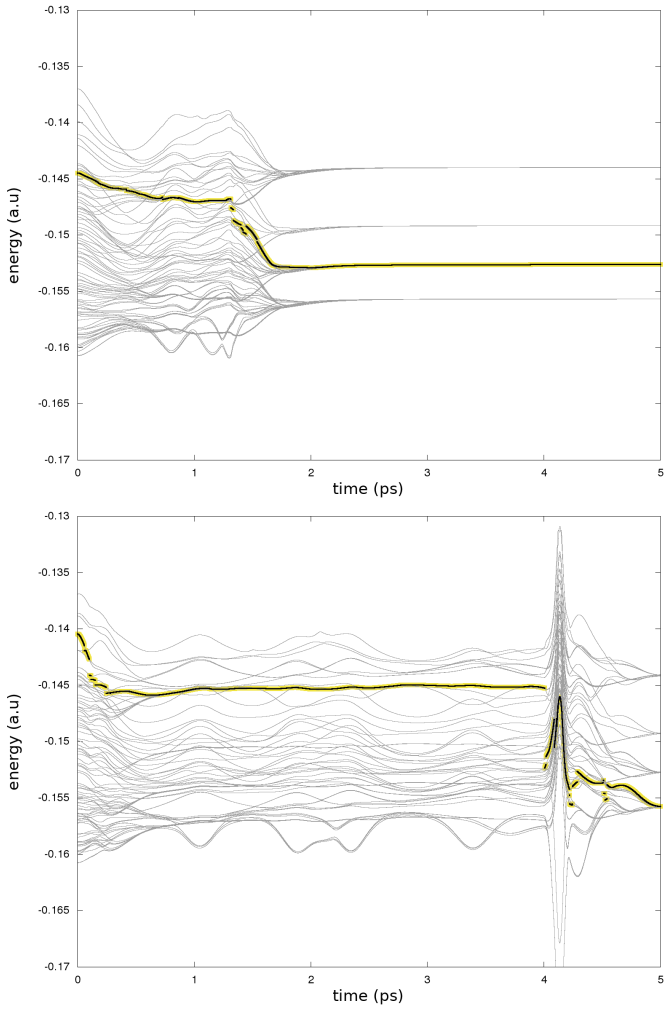



FIG. 3: a) Trajectory (#59) decaying rapidly to the $Ar_*$ dissociation limit with J = 1. b) Long lived trajectory (#28) on cluster of atoms before leading to $Ar^*$, J = 2 dissociation limit.

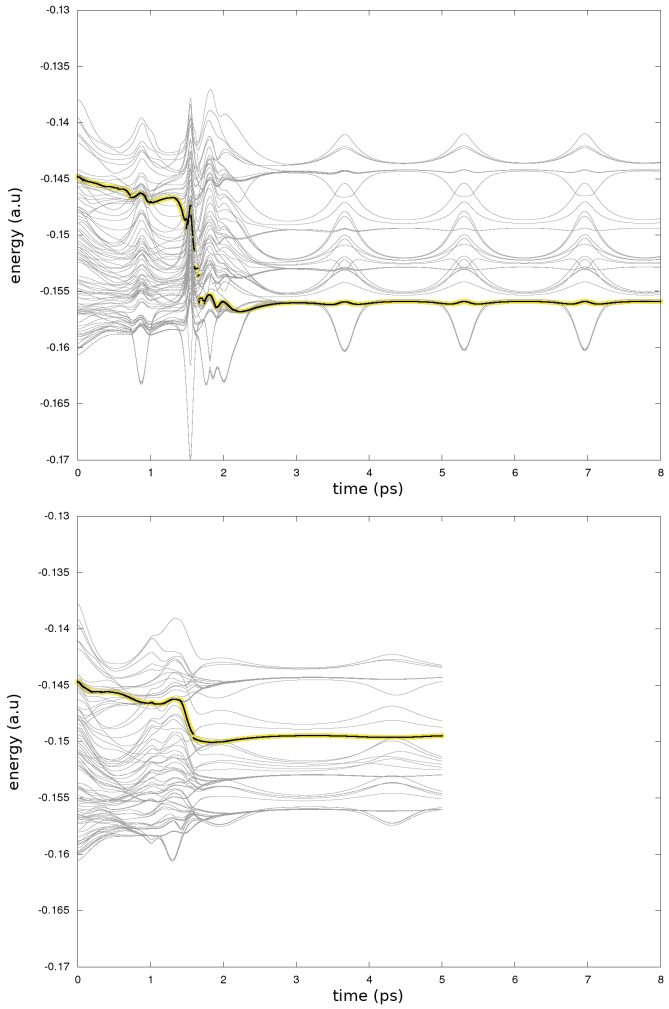



FIG. 4: a) The trajectory (#6) on the PEC leading to the formation of a slowly oscillating ground state dimer. This dynamics is performed for 10 ps and we show the first 8 picoseconds. *Ad hoc* states are not shown here which are higher in energy. The highlighted curve represents the $4s$ dissociation limit of the excited atom and the corresponding, almost flat, PEC. the final state corresponds to $J = 2$. b) The trajectory (#74) leading to formation of $Ar^* + Ar_2 + 4Ar$, with slowly vibrating ground state dimer. This dynamics is performed for 5 ps. *Ad hoc* states are not shown here which are higher in energy. Final state is $J = 0$